Carrier Generation and Recombination
ELEC4603 Wiki Reference

Prepared for Group 13 wiki revision

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Abstract

This document consolidates the wiki material on carrier generation, recombination, carrier lifetime, diffusion length, device applications, and worked sample problems. It is intended as a paste-ready technical reference for OneNote and as a shareable companion to the interactive demonstrations.

Contents

1 Key Definitions

Carrier generation creates mobile electron-hole pairs by promoting electrons from the valence band to the conduction band. Carrier recombination removes electron-hole pairs when conduction-band electrons lose energy and fill valence-band hole states. At thermal equilibrium, microscopic generation and recombination still occur, but their rates balance and the carrier concentrations remain constant [1, 2].

$$n_0 p_0 = n_i^2. \tag{1}$$

When a semiconductor is disturbed by light, voltage bias, heating, or injection,

$$n = n_0 + \Delta n, \qquad p = p_0 + \Delta p. \tag{2}$$

For electron-hole pair generation, the excess carriers are generated together, so $\Delta n = \Delta p$. Throughout this document the sign convention is

$$U = R - G, \tag{3}$$

so $U > 0$ means net recombination and $U < 0$ means net generation.

2 Generation Mechanisms

Generation occurs when energy creates an electron-hole pair. The dominant mechanism depends on temperature, bandgap, illumination, defect density, and electric field strength [1, 2].

Table 1: Important carrier generation mechanisms.

Mechanism	Energy source	Important when	Main device effect
Thermal	Heat / lattice vibrations	High temperature, small bandgap	Leakage and dark current
Optical	Photons	$h\nu \geq E_g$	Photocurrent
Trap-assisted	Defect levels	Depletion regions and poor material	Reverse leakage
Impact ionisation	High electric field	Large reverse bias	Avalanche multiplication

2.1 Thermal Generation

Thermal generation occurs because the lattice has thermal energy at finite temperature:

$$\text{thermal energy} \to e^- + h^+. \tag{4}$$

It is strongly connected to intrinsic carrier concentration,

$$n_i \propto T^{3/2} \exp\left(-\frac{E_g}{2k_\mathrm{B}T}\right). \tag{5}$$

Higher temperature increases thermal generation, while a larger bandgap reduces it. Thermal generation is often unwanted because it contributes to reverse leakage in diodes and dark current in photodiodes and image sensors [3, 4].

2.2 Optical Generation

Optical generation occurs when an absorbed photon has enough energy to excite an electron across the bandgap:

$$h\nu \geq E_g, \qquad \frac{hc}{\lambda} \geq E_g. \tag{6}$$

If the photon energy is below the bandgap, ideal band-to-band absorption cannot create a mobile electron-hole pair. If photon energy is above the bandgap, absorption can generate a mobile electron and hole [1, 5].

For a photon flux Φ_0, a simple depth-dependent generation model is

$$\Phi(x) = \Phi_0 e^{-\alpha x}, \qquad G_{\text{opt}}(x) = \eta\alpha\Phi_0 e^{-\alpha x}. \tag{7}$$

If the incident optical intensity is I_0, then $\Phi_0 \approx I_0/(h\nu)$, so

$$G_{\text{opt}}(x) = \eta\alpha\frac{I_0}{h\nu}e^{-\alpha x}. \tag{8}$$

Large absorption coefficient α means near-surface generation; smaller α means deeper penetration.

2.3 Trap-Assisted Generation

Defects, impurities, and interface states create energy levels inside the bandgap. These levels can act as recombination-generation centres. In a depletion region, mobile carrier concentrations are low, so trap emission can generate carriers that are immediately swept apart by the electric field. A useful approximate depletion-region rate is

$$G_{\text{SRH}} \simeq \frac{n_i}{\tau_g}, \tag{9}$$

where τ_g is the generation lifetime [6, 7, 2].

2.4 Impact Ionisation

Impact ionisation occurs when a high electric field accelerates a carrier so strongly that a collision creates an additional electron-hole pair:

$$\text{fast carrier} + \text{valence electron} \to \text{extra electron} - \text{hole pair}. \tag{10}$$

The new carriers can also be accelerated, producing avalanche multiplication. This is harmful in ordinary diodes if uncontrolled, but useful in avalanche photodiodes [1, 2].

3 Recombination Mechanisms

Recombination mechanisms differ mainly by where the released energy goes and whether defects or extra carriers are involved [1, 2].

3.1 Radiative Recombination

Radiative recombination is direct band-to-band recombination:

$$e^- + h^+ \to h\nu. \tag{11}$$

The common net rate is

$$U_{\text{rad}} = B(np - n_i^2), \tag{12}$$

where B is the radiative coefficient. Direct-bandgap materials such as GaAs support efficient radiative recombination because energy and momentum conservation are easier than in indirect silicon [1, 8].

Table 2: Important carrier recombination mechanisms.

Mechanism	Released energy goes to	Dominant when	Effect
Radiative	Photon	Direct-bandgap materials	LEDs, lasers
Shockley-Read-Hall	Lattice through traps	Defects/traps present	Lifetime and leakage loss
Auger	Another carrier	High doping or injection	Efficiency loss
Surface	Interface traps	Poor passivation	Optical and interface losses

3.2 Shockley-Read-Hall Recombination

Shockley-Read-Hall (SRH) recombination occurs through trap states inside the bandgap. The standard net recombination rate is [6, 7, 2]

$$U_{\mathrm{SRH}} = \frac{np - n_i^2}{\tau_{p0}(n+n_1) + \tau_{n0}(p+p_1)}, \tag{13}$$

where

$$n_1 = n_i \exp\left(\frac{E_t - E_i}{k_{\mathrm{B}}T}\right), \qquad p_1 = n_i \exp\left(\frac{E_i - E_t}{k_{\mathrm{B}}T}\right). \tag{14}$$

Midgap traps are usually strong recombination centres because they are well positioned to capture both electrons and holes.

3.3 Auger Recombination

Auger recombination transfers the recombination energy to a third carrier rather than to a photon:

$$U_{\mathrm{Auger}} = (C_n n + C_p p)(np - n_i^2), \tag{15}$$

where C_n and C_p are Auger coefficients. It becomes important under high carrier concentration, high-level injection, or heavy doping [1, 2].

3.4 Surface Recombination

Surface recombination occurs at semiconductor surfaces and interfaces. Dangling bonds and interface traps provide recombination paths similar to SRH recombination. A common surface recombination flux model is

$$R_s = S\Delta n_s, \tag{16}$$

where S is the surface recombination velocity and Δn_s is excess carrier concentration at the surface. A diffusion boundary condition form is

$$D_n \left.\frac{d\Delta n}{dx}\right|_{\mathrm{surface}} = S_n \Delta n_s. \tag{17}$$

3.5 Effective Lifetime

When independent recombination mechanisms act at the same time, their rates add:

$$\frac{1}{\tau_{\mathrm{eff}}} = \frac{1}{\tau_{\mathrm{rad}}} + \frac{1}{\tau_{\mathrm{SRH}}} + \frac{1}{\tau_{\mathrm{Auger}}} + \frac{1}{\tau_{\mathrm{surface}}}. \tag{18}$$

The shortest lifetime usually dominates because it contributes the largest rate.

4 Carrier Lifetime and Diffusion Length

Carrier lifetime τ describes how long excess carriers survive before recombining. Diffusion length L describes how far they travel before recombining. These quantities determine whether generated carriers are collected or lost [5, 11].

4.1 Excess Carrier Decay

Under low-level injection, the net recombination rate for minority electrons in p-type material is

$$U_n \simeq \frac{\Delta n}{\tau_n}. \tag{19}$$

For minority holes in n-type material,

$$U_p \simeq \frac{\Delta p}{\tau_p}. \tag{20}$$

When the excitation is removed, excess carriers decay exponentially:

$$\Delta n(t) = \Delta n(0) e^{-t/\tau_n}. \tag{21}$$

After one lifetime,

$$\Delta n(\tau_n) = \Delta n(0) e^{-1} \simeq 0.368\, \Delta n(0). \tag{22}$$

Under steady uniform generation,

$$\Delta n_{\mathrm{ss}} = G_L \tau_n. \tag{23}$$

4.2 Diffusion Length

The diffusion lengths are

$$L_n = \sqrt{D_n \tau_n}, \qquad L_p = \sqrt{D_p \tau_p}. \tag{24}$$

The Einstein relation connects diffusivity and mobility:

$$D_n = \mu_n \frac{k_{\mathrm{B}} T}{q} = \mu_n V_{\mathrm{T}}, \qquad D_p = \mu_p \frac{k_{\mathrm{B}} T}{q} = \mu_p V_{\mathrm{T}}. \tag{25}$$

A larger diffusion coefficient or longer lifetime gives a longer diffusion length.

Table 3: Why lifetime and diffusion length matter.

Context	Effect
Solar cells	Longer L improves carrier collection.
Photodiodes	Carriers must be collected before recombining.
LEDs	Recombination is desired only when radiative.
Fast diodes	Shorter τ reduces stored minority charge.
Defective material	More traps reduce τ and L.

5 Generation and Recombination in Devices

5.1 P-N Junction Diodes

Forward bias reduces the junction barrier. Majority carriers cross the junction and become minority carriers on the opposite side:

$$\text{carrier injection} \rightarrow \text{minority carrier diffusion} \rightarrow \text{recombination}. \tag{26}$$

Table 4: Useful and unwanted effects in devices.

Device	Useful effect	Unwanted effect
P-N diode	Recombination supports forward current	Generation causes reverse leakage
Solar cell	Light generation creates photocurrent	Recombination reduces collected current
Photodiode	Optical generation creates signal	Thermal generation causes dark current
LED	Radiative recombination emits light	Non-radiative recombination wastes energy
Switching diode	Short lifetime improves turn-off	Too much recombination increases loss
BJT	Carrier injection enables gain	Base recombination reduces gain

This injected minority-carrier diffusion and recombination supports forward current. In reverse bias, thermal and trap-assisted generation in or near the depletion region is swept out by the electric field, producing leakage [9, 10].

For a long-base diode, an ideal diffusion-limited saturation current is

$$I_S = qAn_i^2 \left(\frac{D_p}{L_p N_D} + \frac{D_n}{L_n N_A} \right). \tag{27}$$

5.2 Solar Cells and Photodiodes

In solar cells and photodiodes,

$$\text{light} \to \text{electron} - \text{hole pairs} \to \text{separated/collected current}. \tag{28}$$

Good performance requires high optical generation, low recombination, long lifetime, long diffusion length, and low surface recombination. Thermal and trap-assisted generation cause unwanted dark current.

5.3 LEDs, Fast Diodes, and Avalanche Devices

LEDs require strong radiative recombination and low non-radiative recombination. Fast switching diodes benefit from shorter lifetime because stored charge scales approximately as

$$Q_{\text{stored}} \propto I_F \tau. \tag{29}$$

Avalanche devices use high-field impact ionisation. In ordinary diodes this can be destructive; in avalanche photodiodes it provides gain.

6 Worked Sample Problems

6.1 Problem 1: Excess Carrier Decay

Question

A silicon sample has $\Delta n(0) = 1.0 \times 10^{15}\,\text{cm}^{-3}$ and $\tau_n = 5\,\mu\text{s}$. After light is switched off, find Δn after $5\,\mu\text{s}$ and the time for Δn to fall to 1% of its initial value.

Using $\Delta n(t) = \Delta n(0)e^{-t/\tau_n}$,

$$\Delta n(5\,\mu\text{s}) = 10^{15}e^{-5/5} = 3.68 \times 10^{14}\,\text{cm}^{-3}. \tag{30}$$

For 1% remaining,

$$e^{-t/\tau_n} = 0.01, \tag{31}$$

$$t = \tau_n \ln(100) = 5(4.605) = 23.0\,\mu\text{s}. \tag{32}$$

6.2 Problem 2: Diffusion Length and Recombination Rate

Question

A p-type silicon sample at 300 K has $N_A = 1.0 \times 10^{16}\,\text{cm}^{-3}$, $n_i = 1.0 \times 10^{10}\,\text{cm}^{-3}$, $D_n = 35\,\text{cm}^2/\text{s}$, $\tau_n = 5\,\mu\text{s}$, and $\Delta n = 2.0 \times 10^{14}\,\text{cm}^{-3}$. Find n_0, L_n, and U_n.

For p-type material, $p_0 \simeq N_A$:

$$n_0 = \frac{n_i^2}{p_0} = \frac{(10^{10})^2}{10^{16}} = 1.0 \times 10^4\,\text{cm}^{-3}. \tag{33}$$

The diffusion length is

$$L_n = \sqrt{D_n\tau_n} = \sqrt{35(5.0 \times 10^{-6})} = 1.32 \times 10^{-2}\,\text{cm} = 132\,\mu\text{m}. \tag{34}$$

The recombination rate is

$$U_n \simeq \frac{\Delta n}{\tau_n} = \frac{2.0 \times 10^{14}}{5.0 \times 10^{-6}} = 4.0 \times 10^{19}\,\text{cm}^{-3}\,\text{s}^{-1}. \tag{35}$$

6.3 Problem 3: Reverse Saturation Current

Question

A long-base silicon diode at 300 K has $A = 1.0 \times 10^{-4}\,\text{cm}^2$, $n_i = 1.0 \times 10^{10}\,\text{cm}^{-3}$, $N_D = N_A = 1.0 \times 10^{16}\,\text{cm}^{-3}$, $D_p = 12\,\text{cm}^2/\text{s}$, $D_n = 35\,\text{cm}^2/\text{s}$, $L_p = 50\,\mu\text{m}$, and $L_n = 100\,\mu\text{m}$. Estimate I_S.

Use

$$I_S = qAn_i^2\left(\frac{D_p}{L_pN_D} + \frac{D_n}{L_nN_A}\right). \tag{36}$$

Convert lengths: $L_p = 5.0 \times 10^{-3}\,\text{cm}$, $L_n = 1.0 \times 10^{-2}\,\text{cm}$. Then

$$\frac{D_p}{L_pN_D} = \frac{12}{(5.0 \times 10^{-3})(10^{16})} = 2.4 \times 10^{-13}, \tag{37}$$

$$\frac{D_n}{L_nN_A} = \frac{35}{(1.0 \times 10^{-2})(10^{16})} = 3.5 \times 10^{-13}. \tag{38}$$

Since $qAn_i^2 = (1.60 \times 10^{-19})(10^{-4})(10^{20}) = 1.60 \times 10^{-3}$,

$$I_S = (1.60 \times 10^{-3})(5.9 \times 10^{-13}) = 9.44 \times 10^{-16}\,\text{A}. \tag{39}$$

6.4 Problem 4: Design Comparison

A short minority-carrier lifetime is useful in a fast switching diode because stored charge must be removed before turn-off:

$$Q_{\text{stored}} \propto I_F\tau. \tag{40}$$

Shorter τ means less stored charge and faster turn-off. In a solar cell, however,

$$L = \sqrt{D\tau}. \tag{41}$$

Shorter τ reduces diffusion length, so more photogenerated carriers recombine before collection.

6.5 Problem 5: Surface Recombination Design

Question

A thin p-type silicon photodiode has $D_n = 35\,\mathrm{cm^2/s}$, $\tau_{\mathrm{bulk}} = 10\,\mu\mathrm{s}$, $W = 20\,\mu\mathrm{m}$, $S = 1.0 \times 10^4\,\mathrm{cm/s}$, and $G_L = 2.0 \times 10^{20}\,\mathrm{cm^{-3}\,s^{-1}}$. Estimate τ_{eff}, L_{eff}, and Δn_{ss}.

For two recombining surfaces,

$$\frac{1}{\tau_{\mathrm{eff}}} \simeq \frac{1}{\tau_{\mathrm{bulk}}} + \frac{2S}{W}. \tag{42}$$

With $W = 2.0 \times 10^{-3}\,\mathrm{cm}$ and $\tau_{\mathrm{bulk}} = 1.0 \times 10^{-5}\,\mathrm{s}$,

$$\frac{1}{\tau_{\mathrm{eff}}} = 10^5 + \frac{2(10^4)}{2.0 \times 10^{-3}} = 1.01 \times 10^7\ \mathrm{s^{-1}}, \tag{43}$$

so

$$\tau_{\mathrm{eff}} = 99\,\mathrm{ns}. \tag{44}$$

Then

$$L_{\mathrm{eff}} = \sqrt{35(9.90 \times 10^{-8})} = 18.6\,\mu\mathrm{m}, \tag{45}$$

and

$$\Delta n_{\mathrm{ss}} = G_L \tau_{\mathrm{eff}} = (2.0 \times 10^{20})(9.90 \times 10^{-8}) = 1.98 \times 10^{13}\,\mathrm{cm^{-3}}. \tag{46}$$

The bulk lifetime was $10\,\mu\mathrm{s}$, but surface recombination reduces the effective lifetime to about 99 ns, showing why surface passivation is critical.

7 Interactive Demonstrations

Two original standalone HTML demonstrations accompany this document.

- **Lifetime and diffusion length simulator**: varies lifetime, surface recombination, generation rate, temperature, and thickness to show excess-carrier decay and spatial survival.
- **P-N diode and photodiode simulator**: varies temperature, doping, lifetime, diffusion constants, area, and illumination to show I_S, photocurrent, V_{OC}, dark I-V, and illuminated I-V.

These can be shared through GitHub Pages or opened locally from the repository.

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